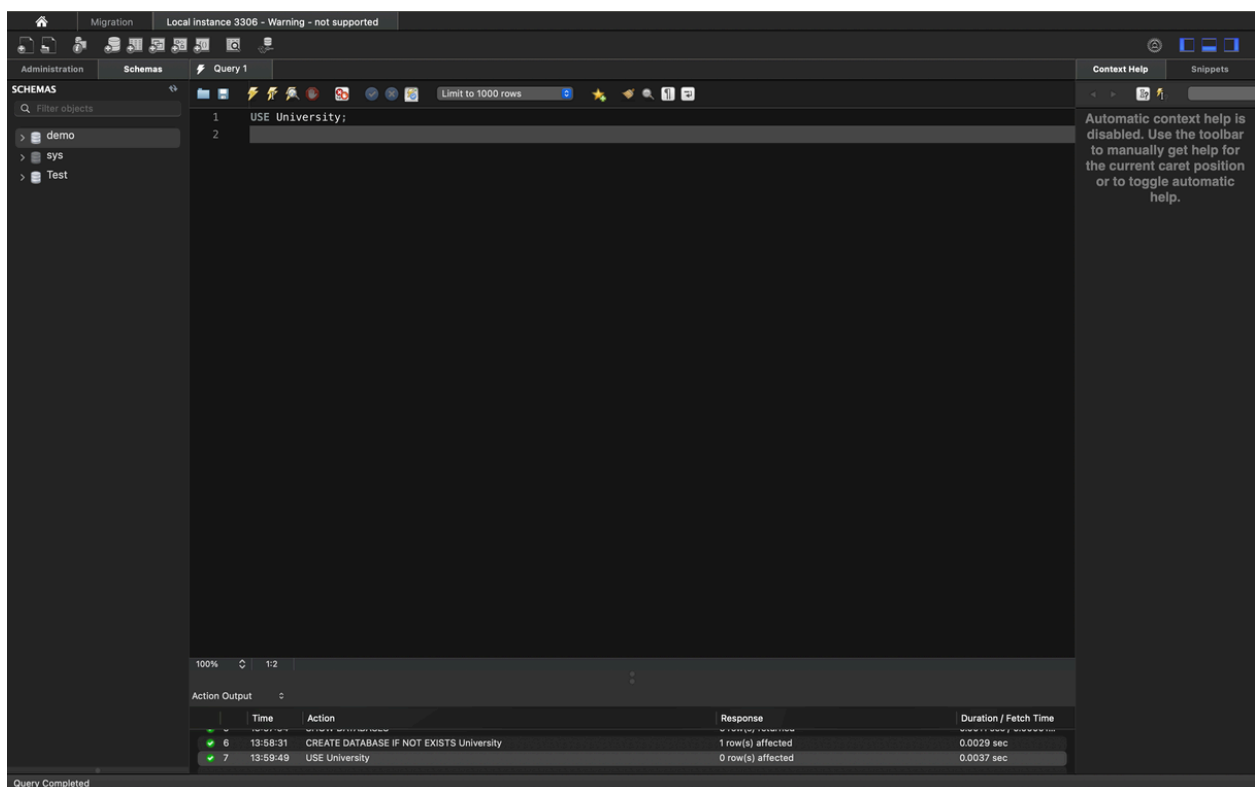
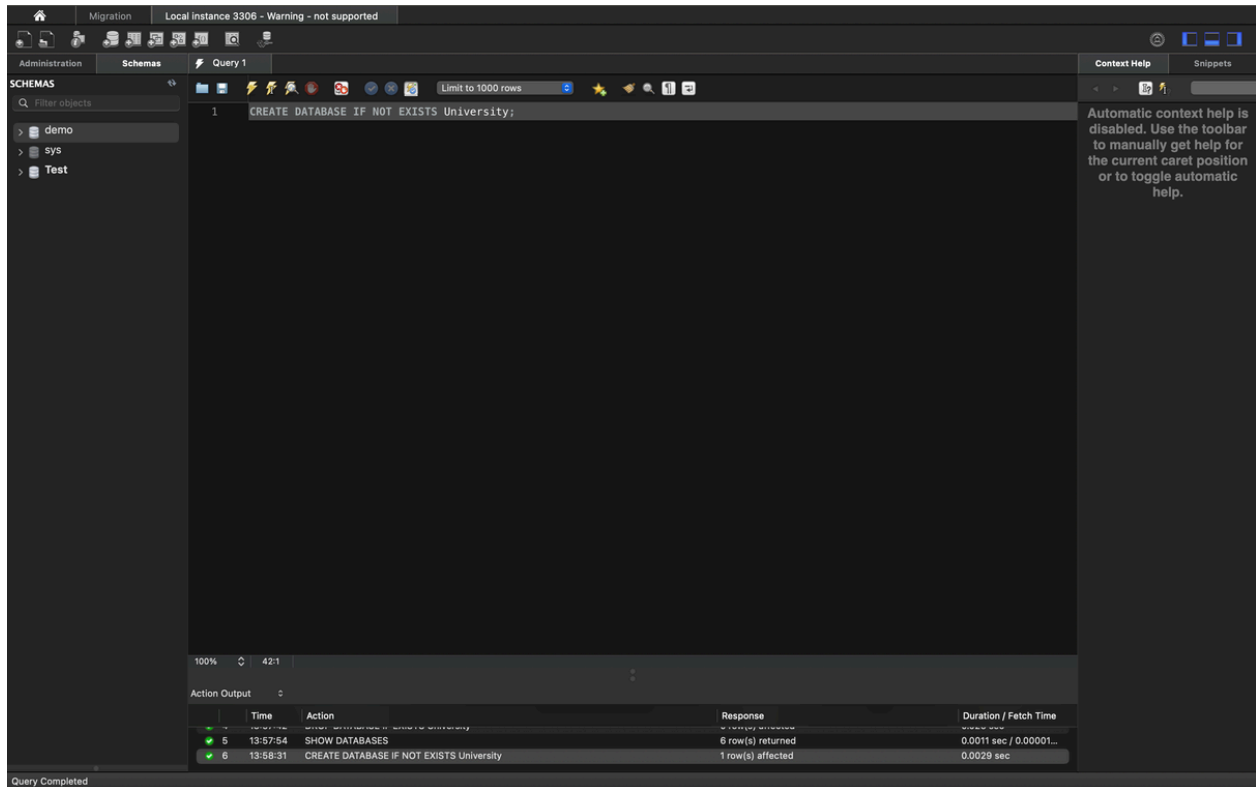


TASK 3 :-

CREATING DATABASE AND USING IT-



CREATING TABLES-

MigrationLocal Instance 3306 - Warning - not supported

AdministrationSchemasQuery 1

Limit to 1000 rows

Context HelpSnippets

SCHEMAS

Filter objects

demo

sys

Test

```
1 CREATE TABLE Department (  
2   dept_id INT PRIMARY KEY,  
3   dept_name VARCHAR(50),  
4   building VARCHAR(50)  
5 );
```

100%3:5

Action Output

	Time	Action	Response	Duration / Fetch Time
7	13:59:49	USE University	0 row(s) affected	0.0037 sec
8	14:00:33	CREATE TABLE Department (dept_id INT PRIMARY KEY, dept_name VARCHAR(50)...	0 row(s) affected	0.070 sec

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MigrationLocal Instance 3306 - Warning - not supported

AdministrationSchemasQuery 1

Limit to 1000 rows

Context HelpSnippets

SCHEMAS

Filter objects

demo

sys

Test

```
1 CREATE TABLE Student (  
2   student_id INT PRIMARY KEY,  
3   name VARCHAR(50),  
4   email VARCHAR(100) UNIQUE,  
5   age INT,  
6   gender VARCHAR(10),  
7   dept_id INT,  
8   FOREIGN KEY (dept_id) REFERENCES Department(dept_id)  
9 );
```

100%3:9

Action Output

	Time	Action	Response	Duration / Fetch Time
1	13:54:41	SELECT CONCAT('DROP DATABASE ', schema_name, ';') FROM information_schema.sche...	2 row(s) returned	0.0046 sec / 0.00001...
2	13:56:51	CREATE DATABASE IF NOT EXISTS University	1 row(s) affected	0.034 sec
3	13:57:14	SHOW DATABASES	7 row(s) returned	0.0072 sec / 0.00003...
4	13:57:42	DROP DATABASE IF EXISTS University	0 row(s) affected	0.026 sec
5	13:57:54	SHOW DATABASES	6 row(s) returned	0.0011 sec / 0.00001...
6	13:58:31	CREATE DATABASE IF NOT EXISTS University	1 row(s) affected	0.0029 sec
7	13:59:49	USE University	0 row(s) affected	0.0037 sec
8	14:00:33	CREATE TABLE Department (dept_id INT PRIMARY KEY, dept_name VARCHAR(50)...	0 row(s) affected	0.070 sec
9	14:01:32	CREATE TABLE Student (student_id INT PRIMARY KEY, name VARCHAR(50), email V...	0 row(s) affected	0.077 sec

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MigrationLocal Instance 3306 - Warning - not supported

AdministrationSchemasQuery 1

Limit to 1000 rows

Context HelpSnippets

SCHEMAS

Filter objects

demo

sys

Test

1

2

3

4

5

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7

8

CREATE TABLE Teacher (
teacher_id INT PRIMARY KEY,
name VARCHAR(50),
email VARCHAR(100) UNIQUE,
designation VARCHAR(50),
dept_id INT,
FOREIGN KEY (dept_id) REFERENCES Department(dept_id)
);

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

100%

3:8

Action Output

	Time	Action	Response	Duration / Fetch Time
✓	13:54:41	SELECT CONCAT('DROP DATABASE ', schema_name, ';') FROM information_schema.sche...	2 row(s) returned	0.0046 sec / 0.00001...
✓	13:56:51	CREATE DATABASE IF NOT EXISTS University	1 row(s) affected	0.034 sec
✓	13:57:14	SHOW DATABASES	7 row(s) returned	0.0072 sec / 0.00003...
✓	13:57:42	DROP DATABASE IF EXISTS University	0 row(s) affected	0.026 sec
✓	13:57:54	SHOW DATABASES	6 row(s) returned	0.0011 sec / 0.00001...
✓	13:58:31	CREATE DATABASE IF NOT EXISTS University	1 row(s) affected	0.0029 sec
✓	13:59:49	USE University	0 row(s) affected	0.0037 sec
✓	14:00:33	CREATE TABLE Department (dept_id INT PRIMARY KEY, dept_name VARCHAR(50),...	0 row(s) affected	0.070 sec
✓	14:01:32	CREATE TABLE Student (student_id INT PRIMARY KEY, name VARCHAR(50), email V...	0 row(s) affected	0.077 sec
✓	14:01:58	CREATE TABLE Teacher (teacher_id INT PRIMARY KEY, name VARCHAR(50), email V...	0 row(s) affected	0.030 sec

Query Completed

MigrationLocal Instance 3306 - Warning - not supported

AdministrationSchemasQuery 1

Limit to 1000 rows

Context HelpSnippets

SCHEMAS

Filter objects

demo

sys

Test

1

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9

CREATE TABLE Course (
course_id INT PRIMARY KEY,
course_name VARCHAR(50),
credits INT,
dept_id INT,
teacher_id INT,
FOREIGN KEY (dept_id) REFERENCES Department(dept_id),
FOREIGN KEY (teacher_id) REFERENCES Teacher(teacher_id)
);

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

100%

3:9

Action Output

	Time	Action	Response	Duration / Fetch Time
✓	13:54:41	SELECT CONCAT('DROP DATABASE ', schema_name, ';') FROM information_schema.sche...	2 row(s) returned	0.0046 sec / 0.00001...
✓	13:56:51	CREATE DATABASE IF NOT EXISTS University	1 row(s) affected	0.034 sec
✓	13:57:14	SHOW DATABASES	7 row(s) returned	0.0072 sec / 0.00003...
✓	13:57:42	DROP DATABASE IF EXISTS University	0 row(s) affected	0.026 sec
✓	13:57:54	SHOW DATABASES	6 row(s) returned	0.0011 sec / 0.00001...
✓	13:58:31	CREATE DATABASE IF NOT EXISTS University	1 row(s) affected	0.0029 sec
✓	13:59:49	USE University	0 row(s) affected	0.0037 sec
✓	14:00:33	CREATE TABLE Department (dept_id INT PRIMARY KEY, dept_name VARCHAR(50),...	0 row(s) affected	0.070 sec
✓	14:01:32	CREATE TABLE Student (student_id INT PRIMARY KEY, name VARCHAR(50), email V...	0 row(s) affected	0.077 sec
✓	14:01:58	CREATE TABLE Teacher (teacher_id INT PRIMARY KEY, name VARCHAR(50), email V...	0 row(s) affected	0.030 sec
✓	14:02:30	CREATE TABLE Course (course_id INT PRIMARY KEY, course_name VARCHAR(50), c...	0 row(s) affected	0.023 sec

Query Completed

MigrationLocal Instance 3306 - Warning - not supported

AdministrationSchemasQuery 1

Context HelpSnippets

SCHEMAS

Filter objects

demo

sys

Test

12

```
1 CREATE TABLE Enrollment (
2   student_id INT,
3   course_id INT,
4   semester INT,
5   grade CHAR(2),
6   PRIMARY KEY (student_id, course_id),
7   FOREIGN KEY (student_id) REFERENCES Student(student_id),
8   FOREIGN KEY (course_id) REFERENCES Course(course_id)
9 );
```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

100%

3:9

Action Output

	Time	Action	Response	Duration / Fetch Time
✓	13:56:51	CREATE DATABASE IF NOT EXISTS University	1 row(s) affected	0.034 sec
✓	13:57:14	SHOW DATABASES	7 row(s) returned	0.0072 sec / 0.00003...
✓	13:57:42	DROP DATABASE IF EXISTS University	0 row(s) affected	0.026 sec
✓	13:57:54	SHOW DATABASES	6 row(s) returned	0.0011 sec / 0.00001...
✓	13:58:31	CREATE DATABASE IF NOT EXISTS University	1 row(s) affected	0.0029 sec
✓	13:59:49	USE University	0 row(s) affected	0.0037 sec
✓	14:00:33	CREATE TABLE Department (dept_id INT PRIMARY KEY, dept_name VARCHAR(50), ...	0 row(s) affected	0.070 sec
✓	14:01:32	CREATE TABLE Student (student_id INT PRIMARY KEY, name VARCHAR(50), email V...	0 row(s) affected	0.077 sec
✓	14:01:58	CREATE TABLE Teacher (teacher_id INT PRIMARY KEY, name VARCHAR(50), email V...	0 row(s) affected	0.030 sec
✓	14:02:30	CREATE TABLE Course (course_id INT PRIMARY KEY, course_name VARCHAR(50), c...	0 row(s) affected	0.023 sec
✓	14:03:00	CREATE TABLE Enrollment (student_id INT, course_id INT, semester INT, grade C...	0 row(s) affected	0.019 sec

Query Completed

MigrationLocal Instance 3306 - Warning - not supported

AdministrationSchemasQuery 1

Context HelpSnippets

SCHEMAS

Filter objects

demo

sys

Test

12

```
1 CREATE TABLE Exam (
2   exam_id INT PRIMARY KEY,
3   exam_type VARCHAR(20),
4   course_id INT,
5   FOREIGN KEY (course_id) REFERENCES Course(course_id)
6 );
```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

100%

3:6

Action Output

	Time	Action	Response	Duration / Fetch Time
✓	13:57:14	SHOW DATABASES	7 row(s) returned	0.0072 sec / 0.00003...
✓	13:57:42	DROP DATABASE IF EXISTS University	0 row(s) affected	0.026 sec
✓	13:57:54	SHOW DATABASES	6 row(s) returned	0.0011 sec / 0.00001...
✓	13:58:31	CREATE DATABASE IF NOT EXISTS University	1 row(s) affected	0.0029 sec
✓	13:59:49	USE University	0 row(s) affected	0.0037 sec
✓	14:00:33	CREATE TABLE Department (dept_id INT PRIMARY KEY, dept_name VARCHAR(50), ...	0 row(s) affected	0.070 sec
✓	14:01:32	CREATE TABLE Student (student_id INT PRIMARY KEY, name VARCHAR(50), email V...	0 row(s) affected	0.077 sec
✓	14:01:58	CREATE TABLE Teacher (teacher_id INT PRIMARY KEY, name VARCHAR(50), email V...	0 row(s) affected	0.030 sec
✓	14:02:30	CREATE TABLE Course (course_id INT PRIMARY KEY, course_name VARCHAR(50), c...	0 row(s) affected	0.023 sec
✓	14:03:00	CREATE TABLE Enrollment (student_id INT, course_id INT, semester INT, grade C...	0 row(s) affected	0.019 sec
✓	14:03:30	CREATE TABLE Exam (exam_id INT PRIMARY KEY, exam_type VARCHAR(20), course...	0 row(s) affected	0.023 sec

Query Completed

Migration Local Instance 3306 - Warning - not supported

Administration Schemas Query 1

SCHEMAS

Filter objects

- demo
- sys
- Test

```
1 CREATE TABLE Result (
2     student_id INT,
3     exam_id INT,
4     marks INT,
5     PRIMARY KEY (student_id, exam_id),
6     FOREIGN KEY (student_id) REFERENCES Student(student_id),
7     FOREIGN KEY (exam_id) REFERENCES Exam(exam_id)
8 );
```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

100% 3:8

Action Output

	Time	Action	Response	Duration / Fetch Time
✓	4 13:57:42	DROP DATABASE IF EXISTS University	0 row(s) affected	0.026 sec
✓	5 13:57:54	SHOW DATABASES	6 row(s) returned	0.0011 sec / 0.00001...
✓	6 13:58:31	CREATE DATABASE IF NOT EXISTS University	1 row(s) affected	0.0029 sec
✓	7 13:59:49	USE University	0 row(s) affected	0.0037 sec
✓	8 14:00:33	CREATE TABLE Department (dept_id INT PRIMARY KEY, dept_name VARCHAR(50)...	0 row(s) affected	0.070 sec
✓	9 14:01:32	CREATE TABLE Student (student_id INT PRIMARY KEY, name VARCHAR(50), email V...	0 row(s) affected	0.077 sec
✓	10 14:01:58	CREATE TABLE Teacher (teacher_id INT PRIMARY KEY, name VARCHAR(50), email V...	0 row(s) affected	0.030 sec
✓	11 14:02:30	CREATE TABLE Course (course_id INT PRIMARY KEY, course_name VARCHAR(50), c...	0 row(s) affected	0.023 sec
✓	12 14:03:00	CREATE TABLE Enrollment (student_id INT, course_id INT, semester INT, grade C...	0 row(s) affected	0.019 sec
✓	13 14:03:30	CREATE TABLE Exam (exam_id INT PRIMARY KEY, exam_type VARCHAR(20), course...	0 row(s) affected	0.023 sec
✓	14 14:03:55	CREATE TABLE Result (student_id INT, exam_id INT, marks INT, PRIMARY KEY st...	0 row(s) affected	0.053 sec

Query Completed

Migration Local Instance 3306 - Warning - not supported

Administration Schemas Query 1

SCHEMAS

Filter objects

- demo
- sys
- Test

```
1 CREATE TABLE 'Library' (
2     library_id INT PRIMARY KEY,
3     book_name VARCHAR(100),
4     author VARCHAR(50),
5     category VARCHAR(50),
6     available_copies INT
7 );
```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

100% 3:7

Action Output

	Time	Action	Response	Duration / Fetch Time
✓	1 13:54:41	SELECT CONCAT('DROP DATABASE ', schema_name, ';') FROM information_schema...	2 row(s) returned	0.0046 sec / 0.00001...
✓	2 13:56:51	CREATE DATABASE IF NOT EXISTS University	1 row(s) affected	0.034 sec
✓	3 13:57:14	SHOW DATABASES	7 row(s) returned	0.0072 sec / 0.00003...
✓	4 13:57:42	DROP DATABASE IF EXISTS University	0 row(s) affected	0.026 sec
✓	5 13:57:54	SHOW DATABASES	6 row(s) returned	0.0011 sec / 0.00001...
✓	6 13:58:31	CREATE DATABASE IF NOT EXISTS University	1 row(s) affected	0.0029 sec
✓	7 13:59:49	USE University	0 row(s) affected	0.0037 sec
✓	8 14:00:33	CREATE TABLE Department (dept_id INT PRIMARY KEY, dept_name VARCHAR(50)...	0 row(s) affected	0.070 sec
✓	9 14:01:32	CREATE TABLE Student (student_id INT PRIMARY KEY, name VARCHAR(50), e...	0 row(s) affected	0.077 sec
✓	10 14:01:58	CREATE TABLE Teacher (teacher_id INT PRIMARY KEY, name VARCHAR(50), e...	0 row(s) affected	0.030 sec
✓	11 14:02:30	CREATE TABLE Course (course_id INT PRIMARY KEY, course_name VARCHAR(50)...	0 row(s) affected	0.023 sec
✓	12 14:03:00	CREATE TABLE Enrollment (student_id INT, course_id INT, semester INT, gra...	0 row(s) affected	0.019 sec
✓	13 14:03:30	CREATE TABLE Exam (exam_id INT PRIMARY KEY, exam_type VARCHAR(20), c...	0 row(s) affected	0.023 sec
✓	14 14:03:55	CREATE TABLE Result (student_id INT, exam_id INT, marks INT, PRIMARY KE...	0 row(s) affected	0.053 sec
✗	15 14:04:19	CREATE TABLE Library (library_id INT PRIMARY KEY, book_name VARCHAR(100)...	Error Code: 1064. You have an error in your SQL syn...	0.0041 sec
✓	16 14:04:56	SHOW TABLES	7 row(s) returned	0.022 sec / 0.00040...
✗	17 14:05:16	CREATE TABLE Library (library_id INT PRIMARY KEY, book_name VARCHAR(100)...	Error Code: 1064. You have an error in your SQL syn...	0.00055 sec
✓	18 14:07:18	CREATE TABLE 'Library' (library_id INT PRIMARY KEY, book_name VARCHAR(100...	0 row(s) affected	0.014 sec

Query Completed

Local instance 3306 - Warning - not supported

Administration Schemas Query 1

SCHEMAS

Filter objects

- demo
- sys
- Test

1 CREATE TABLE Book_Issue (

2 issue_id INT PRIMARY KEY,

3 student_id INT,

4 library_id INT,

5 issue_date DATE,

6 return_date DATE,

7 FOREIGN KEY (student_id) REFERENCES Student(student_id),

8 FOREIGN KEY (library_id) REFERENCES Library(library_id)

9);

Context Help Snippets

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

100% 3:9

Action Output

	Time	Action	Response	Duration / Fetch Time
2	13:56:51	CREATE DATABASE IF NOT EXISTS University	1 row(s) affected	0.034 sec
3	13:57:14	SHOW DATABASES	7 row(s) returned	0.0072 sec / 0.00003...
4	13:57:42	DROP DATABASE IF EXISTS University	0 row(s) affected	0.026 sec
5	13:57:54	SHOW DATABASES	6 row(s) returned	0.0011 sec / 0.00001...
6	13:58:31	CREATE DATABASE IF NOT EXISTS University	1 row(s) affected	0.0029 sec
7	13:59:49	USE University	0 row(s) affected	0.0037 sec
8	14:00:33	CREATE TABLE Department (dept_id INT PRIMARY KEY, dept_name VARCHAR(50)...	0 row(s) affected	0.070 sec
9	14:01:32	CREATE TABLE Student (student_id INT PRIMARY KEY, name VARCHAR(50), e...	0 row(s) affected	0.077 sec
10	14:01:58	CREATE TABLE Teacher (teacher_id INT PRIMARY KEY, name VARCHAR(50), e...	0 row(s) affected	0.030 sec
11	14:02:30	CREATE TABLE Course (course_id INT PRIMARY KEY, course_name VARCHAR(50)...	0 row(s) affected	0.023 sec
12	14:03:00	CREATE TABLE Enrollment (student_id INT, course_id INT, semester INT, gra...	0 row(s) affected	0.019 sec
13	14:03:30	CREATE TABLE Exam (exam_id INT PRIMARY KEY, exam_type VARCHAR(20), c...	0 row(s) affected	0.023 sec
14	14:03:55	CREATE TABLE Result (student_id INT, exam_id INT, marks INT, PRIMARY KE...	0 row(s) affected	0.053 sec
15	14:04:19	CREATE TABLE Library (library_id INT PRIMARY KEY, book_name VARCHAR(100)...	Error Code: 1064. You have an error in your SQL syn...	0.0041 sec
16	14:04:56	SHOW TABLES	7 row(s) returned	0.022 sec / 0.00040...
17	14:05:16	CREATE TABLE Library (library_id INT PRIMARY KEY, book_name VARCHAR(100)...	Error Code: 1064. You have an error in your SQL syn...	0.00055 sec
18	14:07:18	CREATE TABLE Library (library_id INT PRIMARY KEY, book_name VARCHAR(100)...	0 row(s) affected	0.014 sec
19	14:07:48	CREATE TABLE Book_Issue (issue_id INT PRIMARY KEY, student_id INT, library...	Error Code: 1064. You have an error in your SQL syn...	0.0087 sec
20	14:08:00	CREATE TABLE Book_Issue (issue_id INT PRIMARY KEY, student_id INT, library...	Error Code: 1064. You have an error in your SQL syn...	0.00058 sec
21	14:08:08	CREATE TABLE Book_Issue (issue_id INT PRIMARY KEY, student_id INT, librar...	Error Code: 1064. You have an error in your SQL syn...	0.00056 sec
22	14:08:47	CREATE TABLE Book_Issue (issue_id INT PRIMARY KEY, student_id INT, library...	0 row(s) affected	0.060 sec

Query Completed

Limit to 1000 rows

1 SHOW TABLES;

100% 13:1

Result Grid Filter Rows: Search Export:

Tables_in_univers...	
Book_Issue	
Course	
Department	
Enrollment	
Exam	
Library	
Result	
Student	
Teacher	

Result Grid

Form Editor

Field Types

Query Stats

Execution Plan

INSERTING DATA INTO TABLES-

Limit to 1000 rows

1

2

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10

11

INSERT INTO Teacher VALUES

(201,'Dr. Sharma','sharma@gmail.com','Professor',1),

(202,'Dr. Verma','verma@gmail.com','Associate Prof',2),

(203,'Dr. Singh','singh@gmail.com','Professor',3),

(204,'Dr. Mehta','mehta@gmail.com','Assistant Prof',4),

(205,'Dr. Rao','rao@gmail.com','Professor',5),

(206,'Dr. Iyer','iyer@gmail.com','Assistant Prof',6),

(207,'Dr. Khan','khan@gmail.com','Professor',7),

(208,'Dr. Das','das@gmail.com','Associate Prof',8),

(209,'Dr. Roy','roy@gmail.com','Professor',9),

(210,'Dr. Jain','jain@gmail.com','Assistant Prof',10);

100%

55:11

Action Output

	Time	Action	Response	Duration / Fetch Time
18	14:07:18	CREATE TABLE 'Library' (library_id INT PRIMARY KEY, book_name VARCHAR(100...	0 row(s) affected	0.014 sec
19	14:07:48	CREATE TABLE Book_Issue (issue_id INT PRIMARY KEY, student_id INT, library_...	Error Code: 1064. You have an error in your SQL syn...	0.0087 sec
20	14:08:00	CREATE TABLE Book_Issue (issue_id INT PRIMARY KEY, student_id INT, library_...	Error Code: 1064. You have an error in your SQL syn...	0.00058 sec
21	14:08:08	CREATE TABLE 'Book_Issue' (issue_id INT PRIMARY KEY, student_id INT, librar...	Error Code: 1064. You have an error in your SQL syn...	0.00056 sec
22	14:08:47	CREATE TABLE Book_Issue (issue_id INT PRIMARY KEY, student_id INT, library_...	0 row(s) affected	0.060 sec
23	14:09:33	SHOW TABLES;	Error Code: 1064. You have an error in your SQL syn...	0.0047 sec
24	14:09:39	SHOW TABLES	9 row(s) returned	0.047 sec / 0.00039...
25	14:10:39	INSERT INTO Department VALUES (1,'CSE','Block A'), (2,'ECE','Block B'), (3,'EEE','Block...	10 row(s) affected Records: 10 Duplicates: 0 Warni...	0.043 sec
26	14:11:09	INSERT INTO Student VALUES (101,'Rohit','rohit@gmail.com',21,'M',1), (102,'Aman','aman...	10 row(s) affected Records: 10 Duplicates: 0 Warni...	0.046 sec
27	14:11:42	INSERT INTO Teacher VALUES (201,'Dr. Sharma','sharma@gmail.com','Professor',1), (20...	10 row(s) affected Records: 10 Duplicates: 0 Warni...	0.058 sec

Limit to 1000 rows

1

SELECT * FROM Teacher;

100%

22:1

Result Grid

Filter Rows: Search Edit Export/Import

teacher_id	name	email	designation	dept_id
201	Dr. Sharma	sharma@gmail.com	Professor	1
202	Dr. Verma	verma@gmail.com	Associate Prof	2
203	Dr. Singh	singh@gmail.com	Professor	3
204	Dr. Mehta	mehta@gmail.com	Assistant Prof	4
205	Dr. Rao	rao@gmail.com	Professor	5
206	Dr. Iyer	iyer@gmail.com	Assistant Prof	6
207	Dr. Khan	khan@gmail.com	Professor	7
208	Dr. Das	das@gmail.com	Associate Prof	8
209	Dr. Roy	roy@gmail.com	Professor	9
210	Dr. Jain	jain@gmail.com	Assistant Prof	10
NULL	NULL	NULL	NULL	NULL

Result Grid

Form Editor

Field Types

Query Stats

Execution Plan

Teacher 7

Apply

Limit to 1000 rows

1

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9

10

11

INSERT INTO Course VALUES

(301,'DBMS',4,1,201),

(302,'Signals',3,2,202),

(303,'Machines',4,3,203),

(304,'Thermodynamics',3,4,204),

(305,'Structures',4,5,205),

(306,'Networks',3,6,206),

(307,'AI Basics',4,7,207),

(308,'Data Science',4,8,208),

(309,'Biology 101',3,9,209),

(310,'Chemistry 101',3,10,210);

100%

32:11

Action Output

	Time	Action	Response	Duration / Fetch Time
✖	19 14:07:48	CREATE TABLE Book_Issue (issue_id INT PRIMARY KEY, student_id INT, library_...	Error Code: 1064. You have an error in your SQL syn...	0.0087 sec
✖	20 14:08:00	CREATE TABLE Book_Issue (issue_id INT PRIMARY KEY, student_id INT, library_...	Error Code: 1064. You have an error in your SQL syn...	0.00058 sec
✖	21 14:08:08	CREATE TABLE `Book_Issue` (issue_id INT PRIMARY KEY, student_id INT, librar...	Error Code: 1064. You have an error in your SQL syn...	0.00056 sec
✔	22 14:08:47	CREATE TABLE Book_Issue (issue_id INT PRIMARY KEY, student_id INT, library_...	0 row(s) affected	0.060 sec
✖	23 14:09:33	SHOW TABLES:	Error Code: 1064. You have an error in your SQL syn...	0.0047 sec
✔	24 14:09:39	SHOW TABLES	9 row(s) returned	0.047 sec / 0.00039...
✔	25 14:10:39	INSERT INTO Department VALUES (1,'CSE','Block A'), (2,'ECE','Block B'), (3,'EEE','Block...	10 row(s) affected Records: 10 Duplicates: 0 Warni...	0.043 sec
✔	26 14:11:09	INSERT INTO Student VALUES (101,'Rohit','rohit@gmail.com','21','M',1), (102,'Aman','aman...	10 row(s) affected Records: 10 Duplicates: 0 Warni...	0.046 sec
✔	27 14:11:42	INSERT INTO Teacher VALUES (201,'Dr. Sharma','sharma@gmail.com','Professor',1), (20...	10 row(s) affected Records: 10 Duplicates: 0 Warni...	0.058 sec
✔	28 14:12:15	INSERT INTO Course VALUES (301,'DBMS',4,1,201), (302,'Signals',3,2,202), (303,'Machi...	10 row(s) affected Records: 10 Duplicates: 0 Warni...	0.046 sec

Limit to 1000 rows

1

SELECT * FROM Course;

100%

21:11

Result Grid

Filter Rows: Search Edit: Export/Import:

course_id	course_name	credits	dept_id	teacher_id
301	DBMS	4	1	201
302	Signals	3	2	202
303	Machines	4	3	203
304	Thermodynamics	3	4	204
305	Structures	4	5	205
306	Networks	3	6	206
307	AI Basics	4	7	207
308	Data Science	4	8	208
309	Biology 101	3	9	209
310	Chemistry 101	3	10	210
NULL	NULL	NULL	NULL	NULL

Result Grid

Form Editor

Field Types

Query Stats

Execution Plan

Course 8

Apply

```
1  INSERT INTO Enrollment VALUES
2  (101,301,1,'A'),
3  (102,302,1,'B'),
4  (103,301,2,'A'),
5  (104,303,1,'C'),
6  (105,304,2,'B'),
7  (106,305,1,'A'),
8  (107,306,2,'B'),
9  (108,307,1,'A'),
10 (109,308,2,'C'),
11 (110,309,1,'B');
```

100% 17:11

Action Output

	Time	Action	Response	Duration / Fetch Time
✖	20 14:08:00	CREATE TABLE Book_Issue (issue_id INT PRIMARY KEY, student_id INT, library_...	Error Code: 1064. You have an error in your SQL syn...	0.00058 sec
✖	21 14:08:08	CREATE TABLE `Book_Issue` (issue_id INT PRIMARY KEY, student_id INT, librar...	Error Code: 1064. You have an error in your SQL syn...	0.00056 sec
✔	22 14:08:47	CREATE TABLE Book_Issue (issue_id INT PRIMARY KEY, student_id INT, library_...	0 row(s) affected	0.060 sec
✖	23 14:09:33	SHOW TABLES;	Error Code: 1064. You have an error in your SQL syn...	0.0047 sec
✔	24 14:09:39	SHOW TABLES	9 row(s) returned	0.047 sec / 0.00039...
✔	25 14:10:39	INSERT INTO Department VALUES (1,'CSE','Block A'), (2,'ECE','Block B'), (3,'EEE','Block...	10 row(s) affected Records: 10 Duplicates: 0 Warni...	0.043 sec
✔	26 14:11:09	INSERT INTO Student VALUES (101,'Rohit','rohit@gmail.com','21','M','1'), (102,'Aman','aman...	10 row(s) affected Records: 10 Duplicates: 0 Warni...	0.046 sec
✔	27 14:11:42	INSERT INTO Teacher VALUES (201,'Dr. Sharma','sharma@gmail.com','Professor','1'), (20...	10 row(s) affected Records: 10 Duplicates: 0 Warni...	0.058 sec
✔	28 14:12:15	INSERT INTO Course VALUES (301,'DBMS','4,1,201'), (302,'Signals','3,2,202'), (303,'Machi...	10 row(s) affected Records: 10 Duplicates: 0 Warni...	0.046 sec
✔	29 14:12:38	INSERT INTO Enrollment VALUES (101,301,1,'A'), (102,302,1,'B'), (103,301,2,'A'), (104,30...	10 row(s) affected Records: 10 Duplicates: 0 Warni...	0.035 sec

Limit to 1000 rows

```
1 • INSERT INTO Result VALUES
2 (101,401,85),
3 (102,402,78),
4 (103,401,90),
5 (104,403,65),
6 (105,404,70),
7 (106,405,88),
8 (107,406,76),
9 (108,407,92),
10 (109,408,60),
11 (110,409,75);
```

100% 14:11

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 22	14:08:47	CREATE TABLE Book_Issue (issue_id INT PRIMARY KEY, student_id INT, library_...	0 row(s) affected	0.060 sec
✗ 23	14:09:33	SHOW TABLES:	Error Code: 1064. You have an error in your SQL syn...	0.0047 sec
✓ 24	14:09:39	SHOW TABLES	9 row(s) returned	0.047 sec / 0.00039...
✓ 25	14:10:39	INSERT INTO Department VALUES (1,'CSE','Block A'), (2,'ECE','Block B'), (3,'EEE','Block...	10 row(s) affected Records: 10 Duplicates: 0 Warni...	0.043 sec
✓ 26	14:11:09	INSERT INTO Student VALUES (101,'Rohit','rohit@gmail.com','21','M','1'), (102,'Aman','aman...	10 row(s) affected Records: 10 Duplicates: 0 Warni...	0.046 sec
✓ 27	14:11:42	INSERT INTO Teacher VALUES (201,'Dr. Sharma','sharma@gmail.com','Professor','1'), (20...	10 row(s) affected Records: 10 Duplicates: 0 Warni...	0.058 sec
✓ 28	14:12:15	INSERT INTO Course VALUES (301,'DBMS','4,1,201'), (302,'Signals','3,2,202'), (303,'Machi...	10 row(s) affected Records: 10 Duplicates: 0 Warni...	0.046 sec
✓ 29	14:12:38	INSERT INTO Enrollment VALUES (101,301,1,'A'), (102,302,1,'B'), (103,301,2,'A'), (104,30...	10 row(s) affected Records: 10 Duplicates: 0 Warni...	0.035 sec
✓ 30	14:13:05	INSERT INTO Exam VALUES (401,'Mid','301'), (402,'End','302'), (403,'Mid','303'), (404,'End'...	10 row(s) affected Records: 10 Duplicates: 0 Warni...	0.033 sec
✓ 31	14:13:26	INSERT INTO Result VALUES (101,401,85), (102,402,78), (103,401,90), (104,403,65), (1...	10 row(s) affected Records: 10 Duplicates: 0 Warni...	0.0086 sec

Limit to 1000 rows

```
1 • SELECT * FROM Result;
```

100% 21:1

Result Grid Filter Rows: Search Edit: Export/Import:

student_...	exam_id	marks
101	401	85
102	402	78
103	401	90
104	403	65
105	404	70
106	405	88
107	406	76
108	407	92
109	408	60
110	409	75
NULL	NULL	NULL

Result 11 Apply

Result Grid Form Editor Field Types Query Stats Execution Plan

Limit to 1000 rows

1

2

3

4

5

6

7

8

9

10

11

```
INSERT INTO `Library` VALUES
(501,'DBMS Book','Korth','CS',5),
(502,'Signals Book','Oppenheim','ECE',4),
(503,'Machines Book','Bimbhra','EEE',6),
(504,'Thermo Book','Cengel','ME',3),
(505,'Structures Book','Hibbeler','CE',5),
(506,'Networks Book','Tanenbaum','IT',4),
(507,'AI Book','Russell','AI',6),
(508,'DS Book','James','DS',5),
(509,'Bio Book','Campbell','Bio',3),
(510,'Chem Book','Morrison','Chem',4);
```

100%

22:1

Action Output

	Time	Action	Response	Duration / Fetch Time
23	14:09:33	SHOW TABLES:	Error Code: 1064. You have an error in your SQL syn...	0.0047 sec
24	14:09:39	SHOW TABLES	9 row(s) returned	0.047 sec / 0.00039...
25	14:10:39	INSERT INTO Department VALUES (1,'CSE','Block A'), (2,'ECE','Block B'), (3,'EEE','Block...	10 row(s) affected Records: 10 Duplicates: 0 Warni...	0.043 sec
26	14:11:09	INSERT INTO Student VALUES (101,'Rohit','rohit@gmail.com','21','M',1), (102,'Aman','aman...	10 row(s) affected Records: 10 Duplicates: 0 Warni...	0.046 sec
27	14:11:42	INSERT INTO Teacher VALUES (201,'Dr. Sharma','sharma@gmail.com','Professor',1), (20...	10 row(s) affected Records: 10 Duplicates: 0 Warni...	0.058 sec
28	14:12:15	INSERT INTO Course VALUES (301,'DBMS',4,1,201), (302,'Signals',3,2,202), (303,'Machi...	10 row(s) affected Records: 10 Duplicates: 0 Warni...	0.046 sec
29	14:12:38	INSERT INTO Enrollment VALUES (101,301,1,'A'), (102,302,1,'B'), (103,301,2,'A'), (104,30...	10 row(s) affected Records: 10 Duplicates: 0 Warni...	0.035 sec
30	14:13:05	INSERT INTO Exam VALUES (401,'Mid',301), (402,'End',302), (403,'Mid',303), (404,'End'...	10 row(s) affected Records: 10 Duplicates: 0 Warni...	0.033 sec
31	14:13:26	INSERT INTO Result VALUES (101,401,85), (102,402,78), (103,401,90), (104,403,65), (1...	10 row(s) affected Records: 10 Duplicates: 0 Warni...	0.0086 sec
32	14:13:57	INSERT INTO `Library` VALUES (501,'DBMS Book','Korth','CS',5), (502,'Signals Book','O...	10 row(s) affected Records: 10 Duplicates: 0 Warni...	0.0068 sec

Limit to 1000 rows

1

```
SELECT * FROM `Library`;
```

100%

23:1

Result Grid

Filter Rows: Search

Edit: Export/Import:

library_id	book_name	author	category	available_copies
501	DBMS Book	Korth	CS	5
502	Signals Book	Oppenheim	ECE	4
503	Machines Book	Bimbhra	EEE	6
504	Thermo Book	Cengel	ME	3
505	Structures Book	Hibbeler	CE	5
506	Networks Book	Tanenbaum	IT	4
507	AI Book	Russell	AI	6
508	DS Book	James	DS	5
509	Bio Book	Campbell	Bio	3
510	Chem Book	Morrison	Chem	4
NULL	NULL	NULL	NULL	NULL

Result Grid

Form Editor

Field Types

Query Stats

Execution Plan

Library 12

Apply

OPERATIONS-

1.INSERTING DATA-

[illegible]

2.UPDATING DATA-

[illegible]

3.DELETING DATA-

The screenshot shows a database management tool interface. The top section contains SQL queries:

```
1 DELETE FROM Book_Issue
2 WHERE library_id = 510;
3
4 DELETE FROM `Library`
5 WHERE library_id = 510;
6 SELECT * FROM `Library`;
```

Below the queries is a "Result Grid" showing the results of the queries. The grid has columns: library_id, book_name, author, category, available_copies. The data is as follows:

library_id	book_name	author	category	available_copies
1	DBMS	Korth	Education	5
2	DBMS	Korth	Education	5
501	DBMS Book	Korth	CS	4
502	Signals Book	Oppenheim	ECE	4
503	Machines Book	Bimbhra	EEE	6
504	Thermo Book	Cengel	ME	3
505	Structures Book	Hibbeler	CE	5
506	Networks Book	Tanenbaum	IT	4
507	AI Book	Russell	AI	6
508	DS Book	James	DS	5
509	Bio Book	Campbell	Bio	3
NULL	NULL	NULL	NULL	NULL

Below the result grid is an "Action Output" table showing the execution of the queries:

Time	Action	Response	Duration / Fetch Time
14:29:05	DELETE FROM `Library` WHERE library_id = 510	1 row(s) affected	0.0038 sec
14:29:05	SELECT * FROM `Library` LIMIT 0, 1000	11 row(s) returned	0.0012 sec / 0.00004...

4.SELECTING BOOKS BY CATEGORY-(WHERE)

The screenshot shows a database management tool interface. The top section contains SQL queries:

```
1 SELECT * FROM `Library`
2 WHERE category = 'CS';
```

Below the queries is a "Result Grid" showing the results of the queries. The grid has columns: library_id, book_name, author, category, available_copies. The data is as follows:

library_id	book_name	author	category	available_copies
501	DBMS Book	Korth	CS	4
NULL	NULL	NULL	NULL	NULL

Below the result grid is an "Action Output" table showing the execution of the queries:

Time	Action	Response	Duration / Fetch Time
14:30:32	SELECT * FROM `Library` WHERE category = 'Education' LIMIT 0, 1000	2 row(s) returned	0.017 sec / 0.00049 s...
14:31:11	SELECT * FROM `Library` WHERE category = 'CS' LIMIT 0, 1000	1 row(s) returned	0.0021 sec / 0.00004...

5.JOIN OPERATION-SHOWING ISSUED BOOKS WITH STUDENT DETAILS(JOIN)-

The screenshot shows a database query editor with a query window and a results grid. The query is a JOIN operation between a student table and a book issue table. The results grid displays 9 rows of data, showing student details and book issue information.

```
1 SELECT s.student_id, s.name, l.book_name, b.issue_date
2 FROM Book_Issue b
3 JOIN Student s ON b.student_id = s.student_id
4 JOIN 'Library' l ON b.library_id = l.library_id;
```

student_id	name	book_name	issue_date
101	Rohit	DBMS Book	2026-03-01
102	Aman	Signals Book	2026-03-02
103	Sheha	Machines Book	2026-03-03
104	Priya	Thermo Book	2026-03-04
105	Karan	Structures Book	2026-03-05
106	Neha	Networks Book	2026-03-06
107	Arjun	AI Book	2026-03-07
108	Pooja	DS Book	2026-03-08
109	Rahul	Bio Book	2026-03-09

Result 20

Action Output

Time	Action	Response	Duration / Fetch Time
14:32:06	SELECT s.student_id, s.student_name, l.book_name, b.issue_date FROM Book_Issue b...	Error Code: 1054. Unknown column 's.student_name' in 'field list'	0.0025 sec
14:32:50	SELECT s.student_id, s.name, l.book_name, b.issue_date FROM Book_Issue b JOIN Stu...	9 row(s) returned	0.0016 sec / 0.00004...

6.COUNTING BOOKS BY CATEGORY(GROUP BY)-

The screenshot shows a database query editor with a query window and a results grid. The query is a GROUP BY operation counting the number of books in each category. The results grid displays 10 rows of data, showing the category and the total number of books.

```
1 SELECT category, COUNT(*) AS total_books
2 FROM 'Library'
3 GROUP BY category;
```

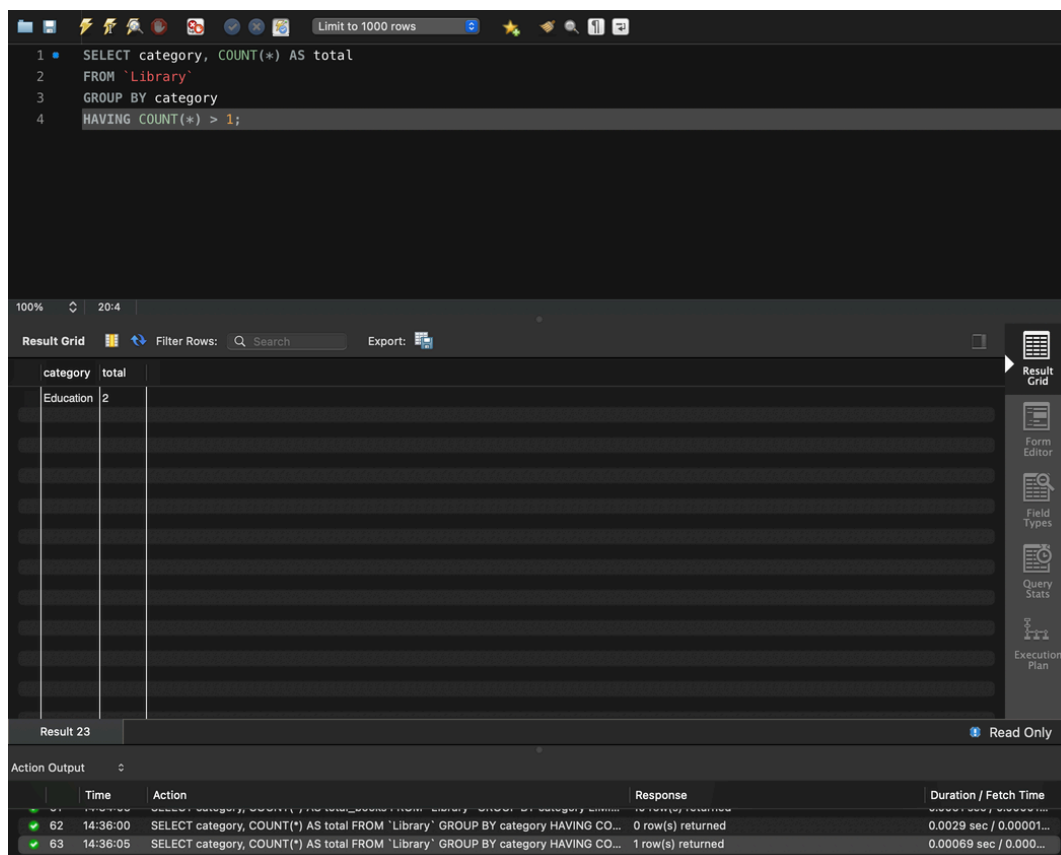
category	total_books
Education	2
CS	1
ECE	1
EEE	1
ME	1
CE	1
IT	1
AI	1
DS	1
Bio	1

Result 21

Action Output

Time	Action	Response	Duration / Fetch Time
14:32:50	SELECT s.student_id, s.name, l.book_name, b.issue_date FROM Book_Issue b JOIN Stu...	9 row(s) returned	0.0016 sec / 0.00004...
14:34:05	SELECT category, COUNT(*) AS total_books FROM 'Library' GROUP BY category LIM...	10 row(s) returned	0.0051 sec / 0.00001...

7.CATEGORIES WITH MORE THAN 1 BOOK(HAVING)-



```

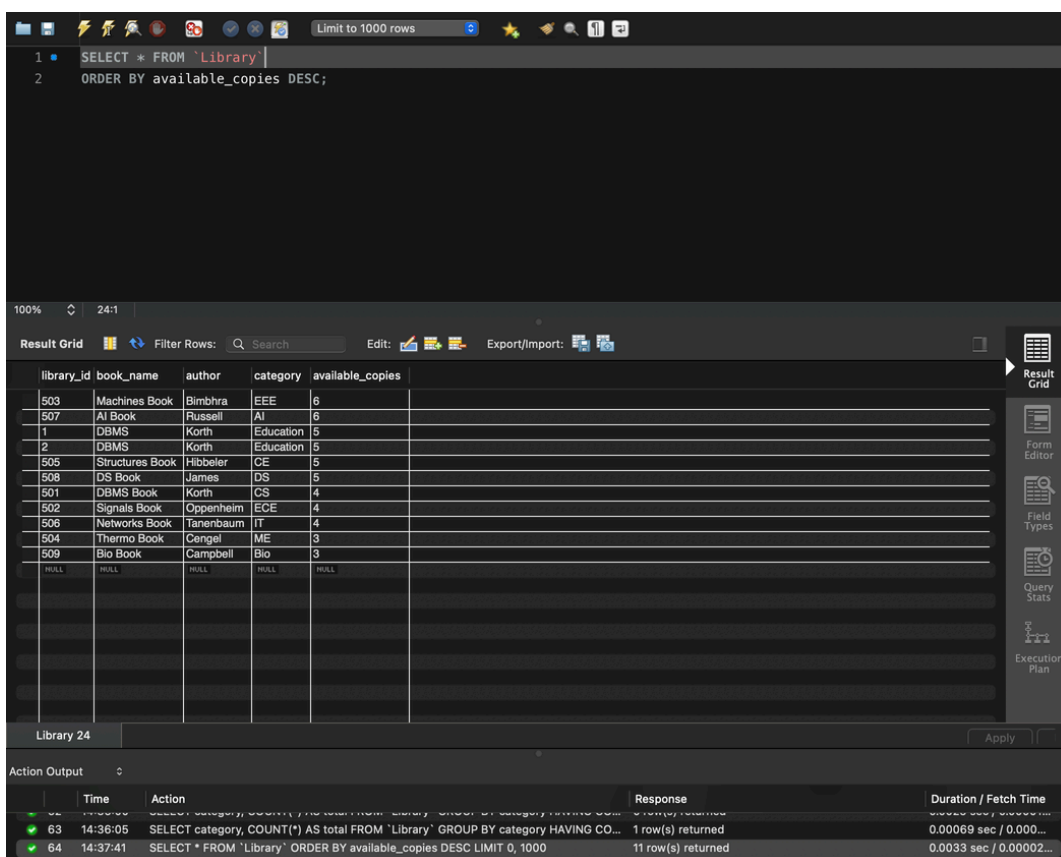
1 • SELECT category, COUNT(*) AS total
2   FROM 'Library'
3   GROUP BY category
4   HAVING COUNT(*) > 1;

```

category	total
Education	2

Time	Action	Response	Duration / Fetch Time
14:36:00	SELECT category, COUNT(*) AS total FROM 'Library' GROUP BY category HAVING CO...	0 row(s) returned	0.0029 sec / 0.00001...
14:36:05	SELECT category, COUNT(*) AS total FROM 'Library' GROUP BY category HAVING CO...	1 row(s) returned	0.00069 sec / 0.000...

8.ORDER BOOKS BY AVAILABLE COPIES(ORDER BY)-



```

1 • SELECT * FROM 'Library'
2   ORDER BY available_copies DESC;

```

library_id	book_name	author	category	available_copies
503	Machines Book	Bimbhra	EEE	6
507	AI Book	Russell	AI	6
1	DBMS	Korth	Education	5
2	DBMS	Korth	Education	5
505	Structures Book	Hibbeler	CE	5
508	DS Book	James	DS	5
501	DBMS Book	Korth	CS	4
502	Signals Book	Oppenheim	ECE	4
506	Networks Book	Tanenbaum	IT	4
504	Thermo Book	Cengel	ME	3
509	Bio Book	Campbell	Bio	3
NULL	NULL	NULL	NULL	NULL

Time	Action	Response	Duration / Fetch Time
14:36:05	SELECT category, COUNT(*) AS total FROM 'Library' GROUP BY category HAVING CO...	1 row(s) returned	0.00069 sec / 0.000...
14:37:41	SELECT * FROM 'Library' ORDER BY available_copies DESC LIMIT 0, 1000	11 row(s) returned	0.0033 sec / 0.00002...

9.STUDENTS WHO ISSUED BOOKS(SUBQUERY)-

Limit to 1000 rows

```

1 SELECT name
2 FROM Student
3 WHERE student_id IN (
4     SELECT student_id FROM Book_Issue
5 );

```

100% 3:5

Result Grid Filter Rows: Search Export:

name
Rohit
Aman
Sneha
Priya
Karan
Neha
Arjun
Pooja
Rahul

Student 25 Read Only

Action Output

	Time	Action	Response	Duration / Fetch Time
64	14:37:41	SELECT * FROM 'Library' ORDER BY available_copies DESC LIMIT 0, 1000	11 row(s) returned	0.0033 sec / 0.00002...
65	14:39:03	SELECT name FROM Student WHERE student_id IN (SELECT student_id FROM Book...	9 row(s) returned	0.0038 sec / 0.00002...

10.SELECTING BOOKS WITH MAXIMUM COPIES(SUBQUERY)-

Limit to 1000 rows

```

1 SELECT book_name, available_copies
2 FROM 'Library'
3 WHERE available_copies = (
4     SELECT MAX(available_copies) FROM 'Library'
5 );

```

100% 48:4

Result Grid Filter Rows: Search Export:

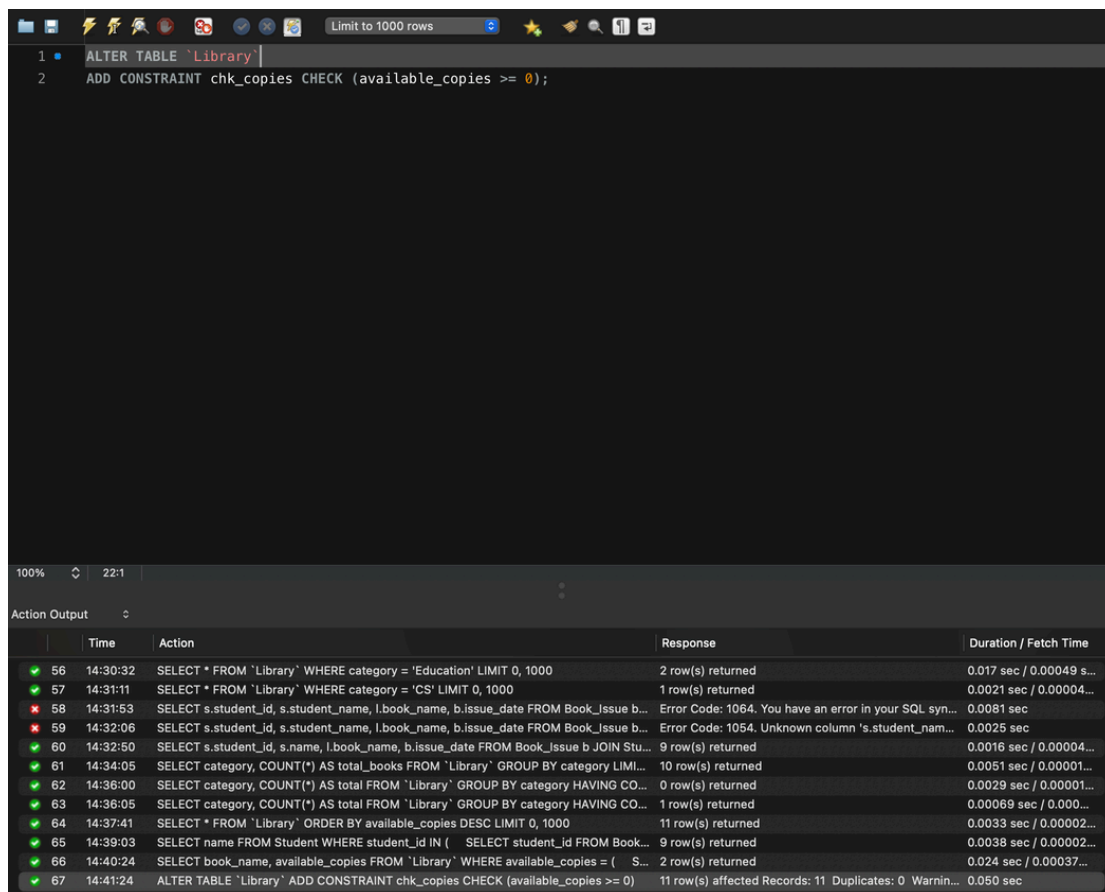
book_name	available_copies
Machines Book	6
AI Book	6

Library 26 Read Only

Action Output

	Time	Action	Response	Duration / Fetch Time
65	14:39:03	SELECT name FROM Student WHERE student_id IN (SELECT student_id FROM Book...	9 row(s) returned	0.0038 sec / 0.00002...
66	14:40:24	SELECT book_name, available_copies FROM 'Library' WHERE available_copies = (S...	2 row(s) returned	0.024 sec / 0.00037...

11.CONSTRAINT-



The screenshot shows a database management tool interface. At the top, there's a toolbar with various icons and a 'Limit to 1000 rows' dropdown. Below the toolbar, the SQL editor contains two commands:

```
1 ALTER TABLE `Library`  
2 ADD CONSTRAINT chk_copies CHECK (available_copies >= 0);
```

Below the editor, the 'Action Output' section displays a table of execution results:

	Time	Action	Response	Duration / Fetch Time
✓ 56	14:30:32	SELECT * FROM `Library` WHERE category = 'Education' LIMIT 0, 1000	2 row(s) returned	0.017 sec / 0.00049 s...
✓ 57	14:31:11	SELECT * FROM `Library` WHERE category = 'CS' LIMIT 0, 1000	1 row(s) returned	0.0021 sec / 0.00004...
✗ 58	14:31:53	SELECT s.student_id, s.student_name, l.book_name, b.issue_date FROM Book_Issue b...	Error Code: 1064. You have an error in your SQL syn...	0.0081 sec
✗ 59	14:32:06	SELECT s.student_id, s.student_name, l.book_name, b.issue_date FROM Book_Issue b...	Error Code: 1054. Unknown column 's.student_nam...	0.0025 sec
✓ 60	14:32:50	SELECT s.student_id, s.name, l.book_name, b.issue_date FROM Book_Issue b JOIN Stu...	9 row(s) returned	0.0016 sec / 0.00004...
✓ 61	14:34:05	SELECT category, COUNT(*) AS total_books FROM `Library` GROUP BY category LIMI...	10 row(s) returned	0.0051 sec / 0.00001...
✓ 62	14:36:00	SELECT category, COUNT(*) AS total FROM `Library` GROUP BY category HAVING CO...	0 row(s) returned	0.0029 sec / 0.00001...
✓ 63	14:36:05	SELECT category, COUNT(*) AS total FROM `Library` GROUP BY category HAVING CO...	1 row(s) returned	0.00069 sec / 0.000...
✓ 64	14:37:41	SELECT * FROM `Library` ORDER BY available_copies DESC LIMIT 0, 1000	11 row(s) returned	0.0033 sec / 0.00002...
✓ 65	14:39:03	SELECT name FROM Student WHERE student_id IN (SELECT student_id FROM Book...	9 row(s) returned	0.0038 sec / 0.00002...
✓ 66	14:40:24	SELECT book_name, available_copies FROM `Library` WHERE available_copies = (S...	2 row(s) returned	0.024 sec / 0.00037...
✓ 67	14:41:24	ALTER TABLE `Library` ADD CONSTRAINT chk_copies CHECK (available_copies >= 0)	11 row(s) affected Records: 11 Duplicates: 0 Warnin...	0.050 sec